



MFU LIBRARY Q&A WEB APPLICATION

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**BACHELOR OF ENGINEERING
IN
COMPUTER ENGINEERING**

**SCHOOL OF INFORMATION TRCHNOLOGY
MAE FAH LUANG UNIVERSITY**

2020

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MFU LIBRARY Q&A WEB APPLICATION

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**A COMPUTER ENGINEERING PROJECT SUBMITTED TO
MAH FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN
COMPUTER ENGINEERING**

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THIS SENIOR PROJECT HAS BE APPROVED
TO BE A PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF BACHELOR OF ENGINEERING
IN
COMPUTER ENGINEERING
2020

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ABSTRACT

This project is made to communicate with Library Staff and MFU Student. This project is Web Application, is called “MFU LIBRARY Q&A WEB APPLICATION”. The function of this web application is about Q&A web application that can help MFU student to understand how to use the library and can communicate with library staff.

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Chapter1

Introduction

1.1 Background and Problem Definition

Every year there will be new students in Mae Fah Luang University Library. It is one of the places that are very necessary for university life. In the use of libraries, there are many rules and methods to use. Many library systems are asked repeatedly over the years.

Web application is an application program that is stored on a remote server and delivered over the Internet through a browser interface. Web services are Web apps by definition and many, although not all, websites contain Web apps.

1.2 Objectives

1.2.1 For communication between users and library staff Reach problems or questions more quickly

1.3 Scope

1.3.1 User

1.3.2 Administration

1.4 Expected Result

1.4.1 Web application can show question and answer for user

1.4.2 Administration can manage question and answer in web application

1.5 Methodology

- 1.5.1 Proposal preparation and defense
- 1.5.2 Literature review and requirement gathering
- 1.5.3 System analysis
- 1.5.4 System design
- 1.5.5 CE Pre-project document preparation
- 1.5.6 CE Pre-project exam
- 1.5.7 System development
- 1.5.8 System testing
- 1.5.9 CE Project document preparation
- 1.5.10 CE Project exam

1.6 Plan

Table 1.1 Working plan

[illegible]

CE Project document preparation									
CE Project exam									

1.7 Equipment, Place and Budget

1.7.1 Place

1.7.1.1 Work at home

1.7.1.2 This project deploys at MFU library server

1.7.2 Device

1.7.2.1 personal laptops.

1.7.2.2 MFU library server

1.7.3 Software

1.7.3.1 IntelliJ

1.7.3.2 visual studio code

1.7.3.3 MySQL database

1.7.3.4 source tree

1.7.3.5 postman

1.7.3.7 Java (Spring boot Framework)

1.7.3.8 TypeScript (angular Framework)

1.7.3.9 CSS

1.7.3.10 JavaScript (Nodejs)

1.7.3.11 HTML

1.7.3.12 Maven

CHAPTER 2

LITERATURE REVIEW

2.1 Theory

A web application is a computer program that uses a web browser to perform a particular function. It is also called a web app. Web apps are present on many websites. A simple example is a contact form on a website.

A web application is a client-server program. It means that it has a client-side and a server-side. The term "client" here refers to the program the individual uses to run the application. It is part of the client-server environment, where many computers share information. For example, in the case of a database, the client is the program through which the user enters data. The server is the application that stores the information.

2.2 Technology and tool

2.2.1 API

An API is a set of programming code that enables data transmission between one software product and another. It also contains the terms of this data exchange.

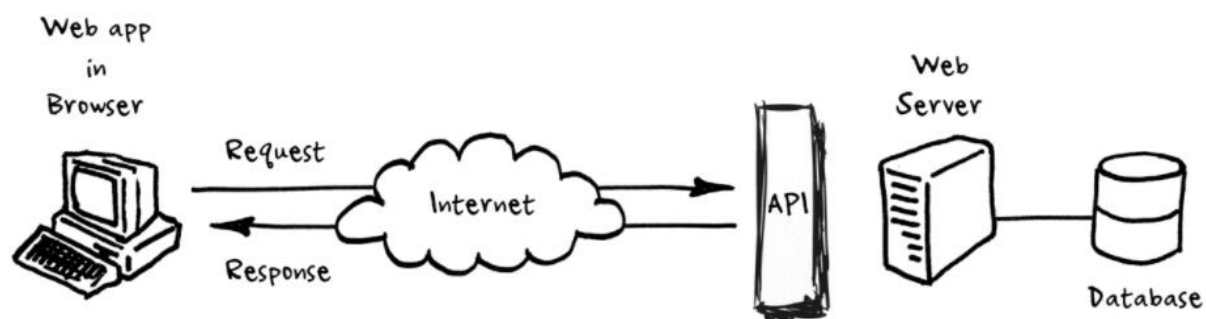


Figure 2.1 API

2.2.2 Framework

Framework is a platform for developing software applications. It provides a foundation on which software developers can build programs for a specific platform. For example, a framework may include predefined classes and functions that can be used to process input, manage hardware devices, and interact with system software. This streamlines the development

process since programmers don't need to reinvent the wheel each time, they develop a new application.

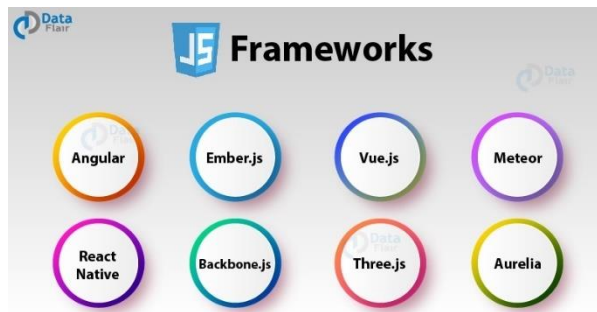


Figure 2.2 Framework

2.2.3 Java (Spring boot framework)

Spring Boot is an open-source Java-based framework used to create a micro-Service. It is developed by Pivotal Team and is used to build stand-alone and production ready spring applications. This chapter will give you an introduction to Spring Boot and familiarizes you with its basic concepts.



Figure 2.3 Spring boot

2.2.4 Maven

Maven is a build automation tool used primarily for Java projects. Maven can also be used to build and manage projects written in C#, Ruby, Scala, and other languages. The Maven

project is hosted by the Apache Software Foundation, where it was formerly part of the Jakarta Project.



Figure 2.4 Maven

2.2.5 Typescript (Angular)

Angular is a TypeScript-based open-source web application framework led by the Angular Team at Google and by a community of individuals and corporations. Angular is a complete rewrite from the same team that built AngularJS.



Figure 2.5 Angular

2.2.6 Postman

Postman is a great tool when trying to dissect RESTful APIs made by others or test ones you have made yourself. It offers a sleek user interface with which to make HTML requests, without the hassle of writing a bunch of code just to test an API's functionality.



Figure 2.6 Postman

2.2.9 visual studio code

Visual Studio Code is a lightweight but powerful source code editor which runs on your desktop and is available for Windows, macOS and Linux. It comes with built-in support for JavaScript, TypeScript and Node.js and has a rich ecosystem of extensions for other languages (such as C++, C#, Java, Python, PHP, Go) and runtimes (such as .NET and Unity).



Figure 2.7 VS Code

2.2.10 IntelliJ

IntelliJ IDEA is an Integrated Development Environment (IDE) for JVM languages designed to maximize developer productivity. It does the routine and repetitive tasks for you by providing clever code completion, static code analysis, and refactoring's, and lets you focus on the bright side of software development, making it not only productive but also an enjoyable experience.

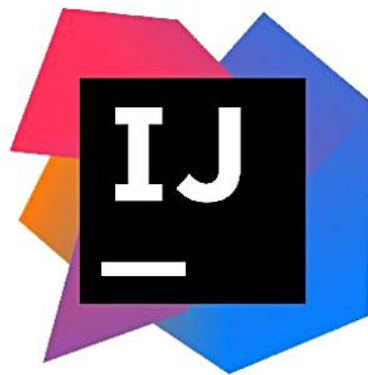


Figure 2.2 IntelliJ

2.2.11 MySQL

MySQL is an open-source relational database management system (RDBMS). A relational database organizes data into one or more data tables in which data types may be related to each other; these relations help structure the data. SQL is a language programmer use to create, modify and extract data from the relational database, as well as control user access to the database. In addition to relational databases and SQL, an RDBMS like MySQL works with an operating system to implement a relational database in a computer's storage system, manages users, allows for network access and facilitates testing database integrity and creation of backups.



Figure 2.2 MySQL

CHAPTER 3

METHODOLOGY

3.1 Use case

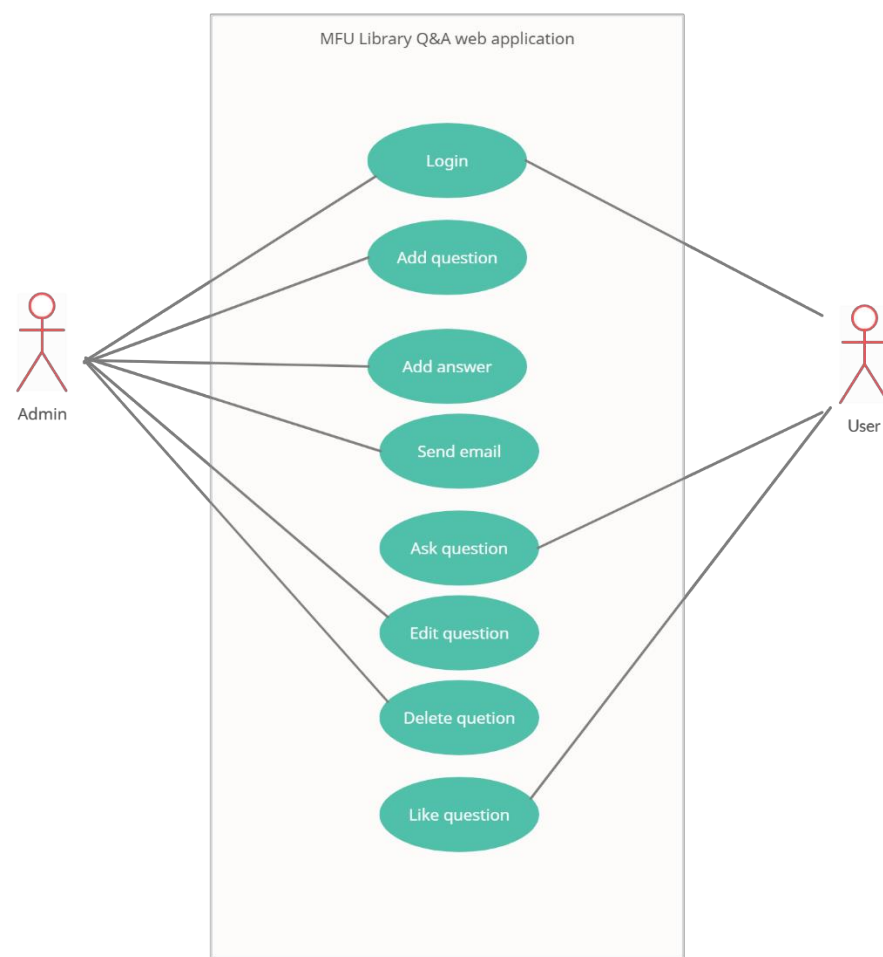


Figure 3.1 Use case

3.2 Database design

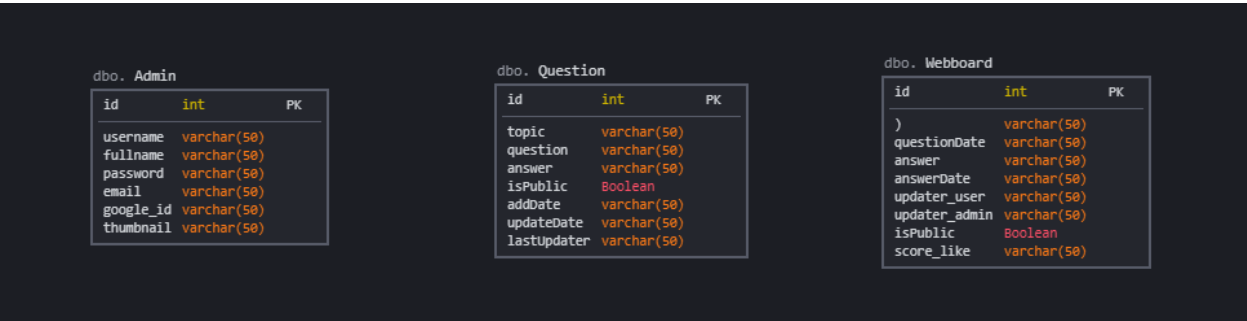


Figure 3.2 Database design

3.3 Web application design

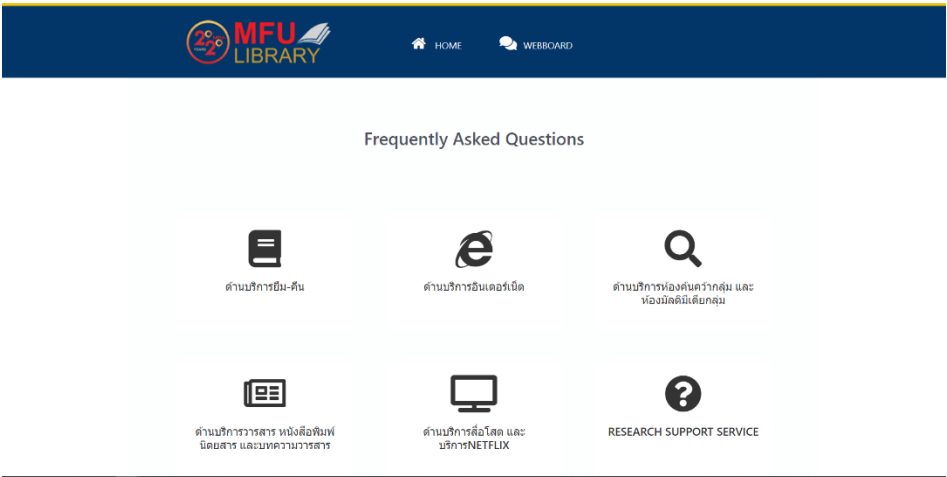


Figure 3.3 Web application design

CHAPTER 4

RESULTS

in this web application, admin can manage question and send E-mail response to user and user can ask question about library.

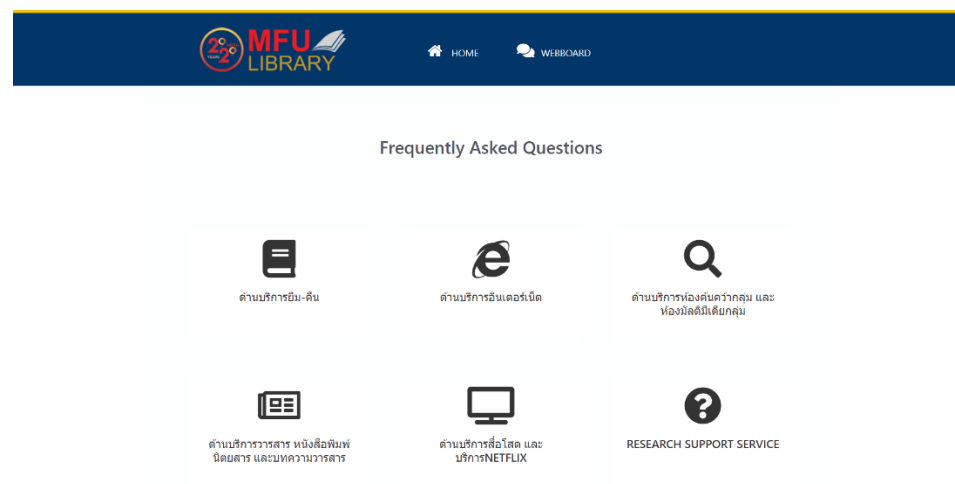


Figure 4.1 Home page

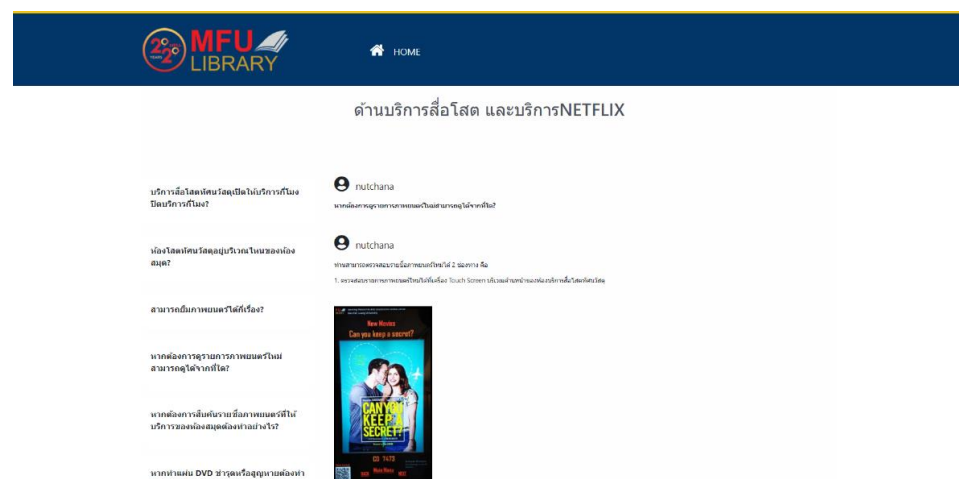


Figure 4.2 Details page

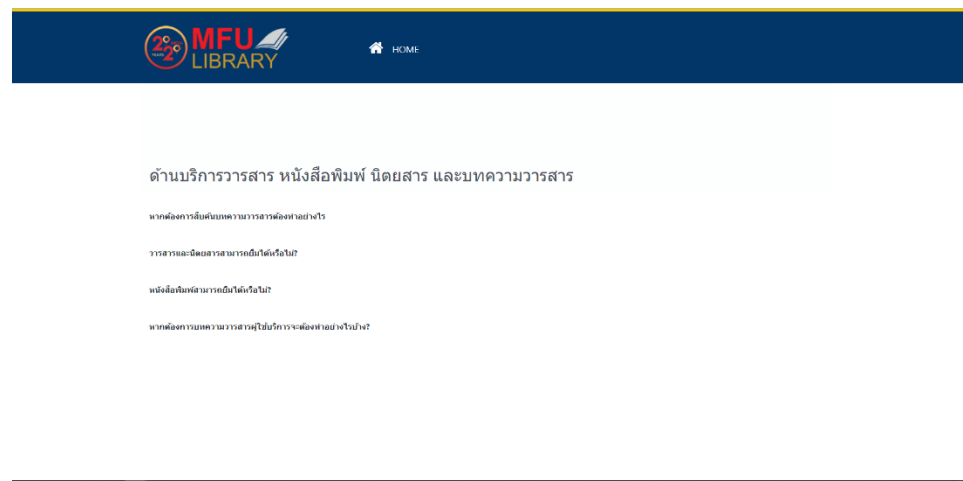


Figure 4.3 Details page

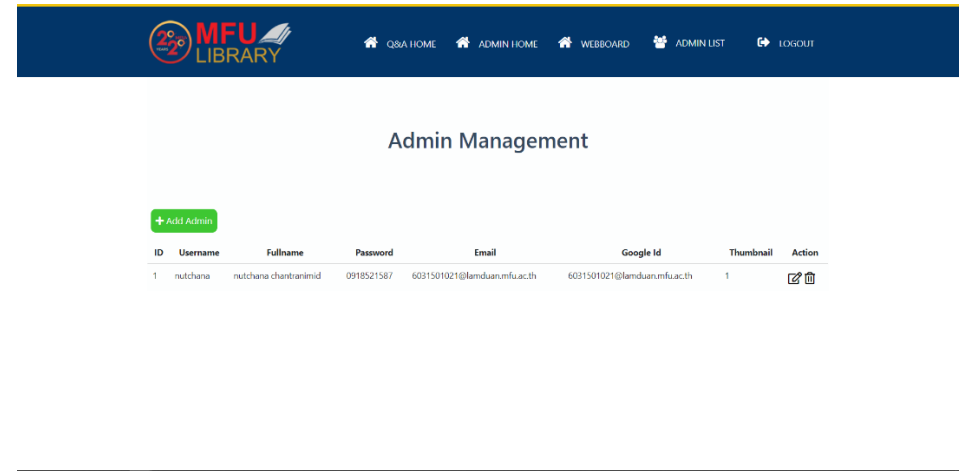


Figure 4.4 Admin page

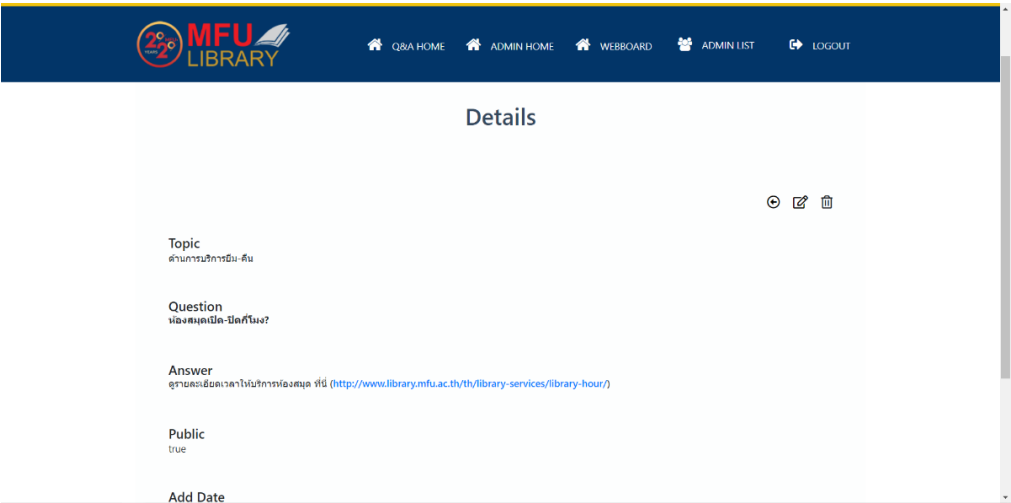


Figure 4.5 Question details page

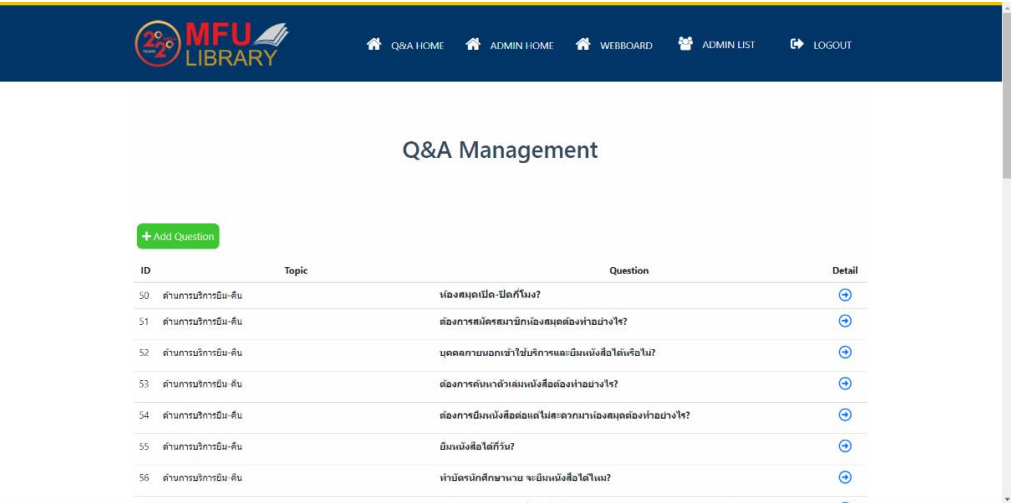


Figure 4.6 Question manage page

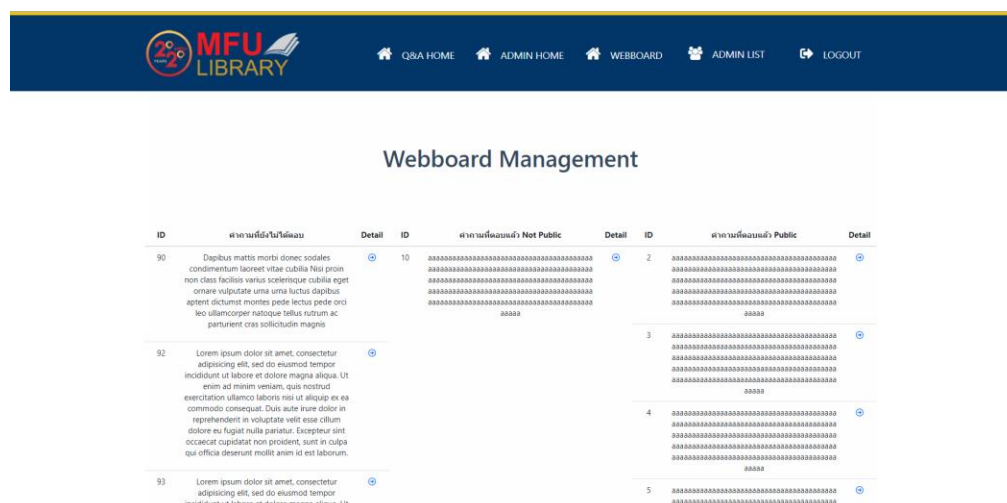


Figure 4.7 Webboard management

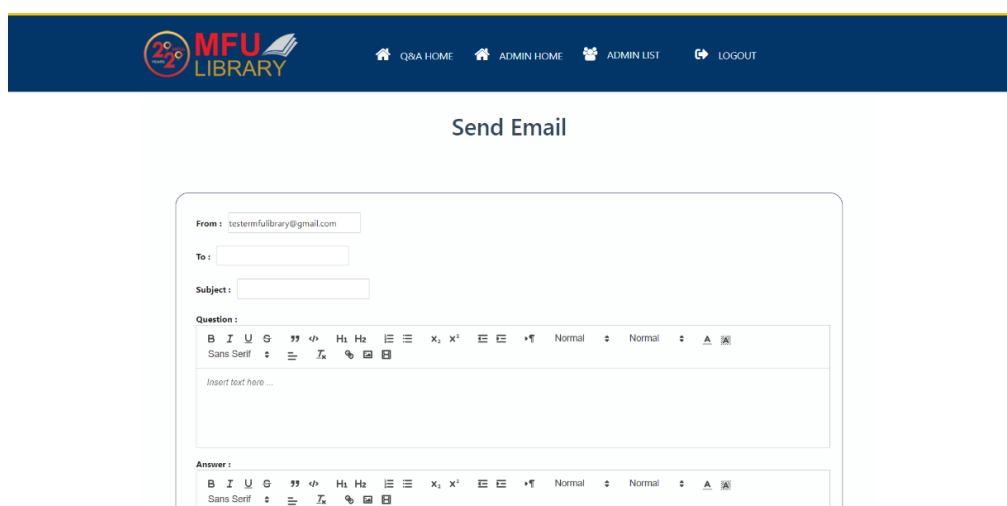


Figure 4.8 Send Email

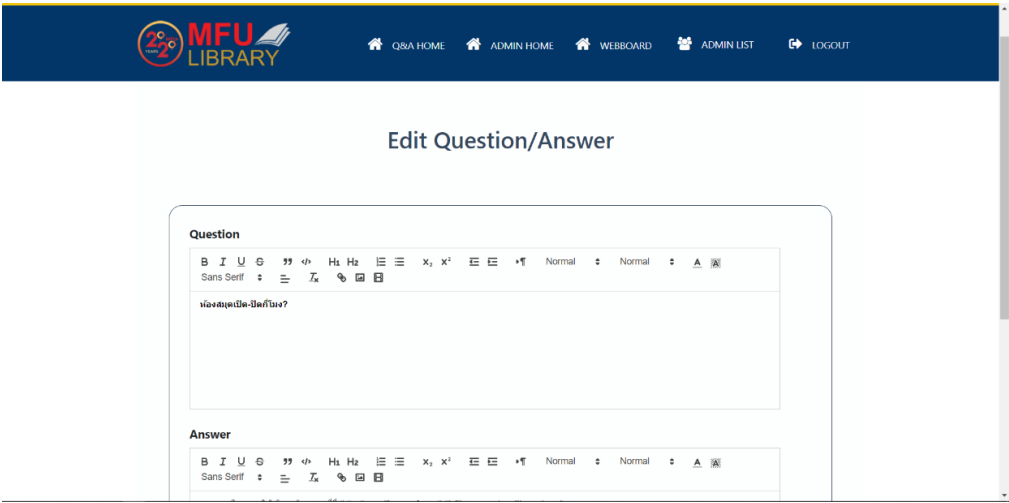


Figure 4.9 Edit Question

CHAPTER 5

CONCLUSION AND DISCUSSION

5.1 Conclusion

MFU library Q&A Web application is the website is dedicated to the high number of repetitive questions, and users can also ask questions about their use in the library, with the library staff responding to them by e-mail.

5.2 Discussion

Result of working in senior projects. Users can log in to the system to inquire about library usage, and administrators can reply via email. Administrators can bring up the FAQ and bring up user-submitted questions, but they have to screen the questions before they're put on the web. Can manage questions such as add, edit, delete.

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APPENDICES

1. Back-end Git

https://gitlab.com/patharapon123/questionandanswer_backend2.git

2. Front-end Git

<https://gitlab.com/nutchana.c/mfulibrary-project-frontend.git>

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